PHARMACOLOGY

Drug-Induced Liver Injury

SGL 4
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What is the difference between intrinsic and idiosyncratic hepatotoxicity, and which one does acetaminophen represent?
2. Which lab value is elevated in hepatocellular injury, and which in cholestatic injury?
3. What is the mechanism linking acetaminophen overdose to hepatocellular necrosis, and how does N-acetylcysteine treat it?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

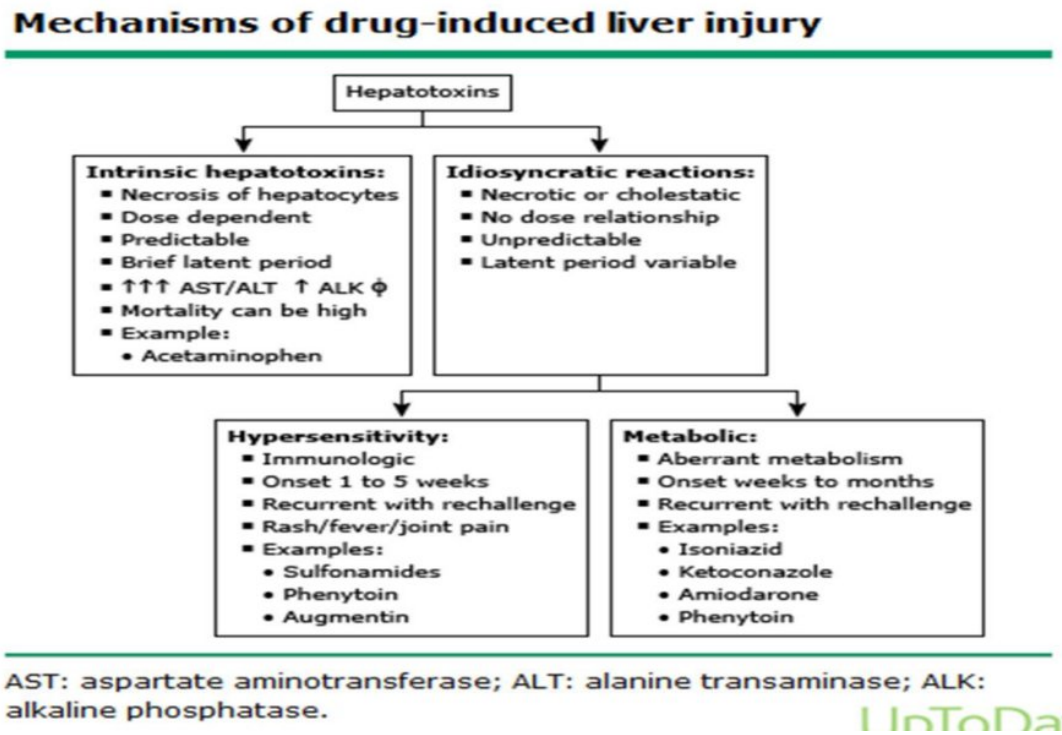
1 DILI — definition, scale & risk factors

Drug-induced liver injury (DILI) is an acute or chronic hepatic response to a drug or compound. Over **1,000 medications/herbals** are known hepatotoxins (LiverTox database, NIDDK). DILI is the **most frequent cause of drug withdrawal** from the market (up to 32% of withdrawals).

Risk factor	Detail
Age	Adults > children
Sex	Females > males
Race/ethnicity	African Americans: predominantly trimethoprim-sulfamethoxazole. White Americans: predominantly amoxicillin-clavulanate
Other	Obesity, diabetes, chronic viral hepatitis
Genetic	Polymorphisms in CYP isoenzymes

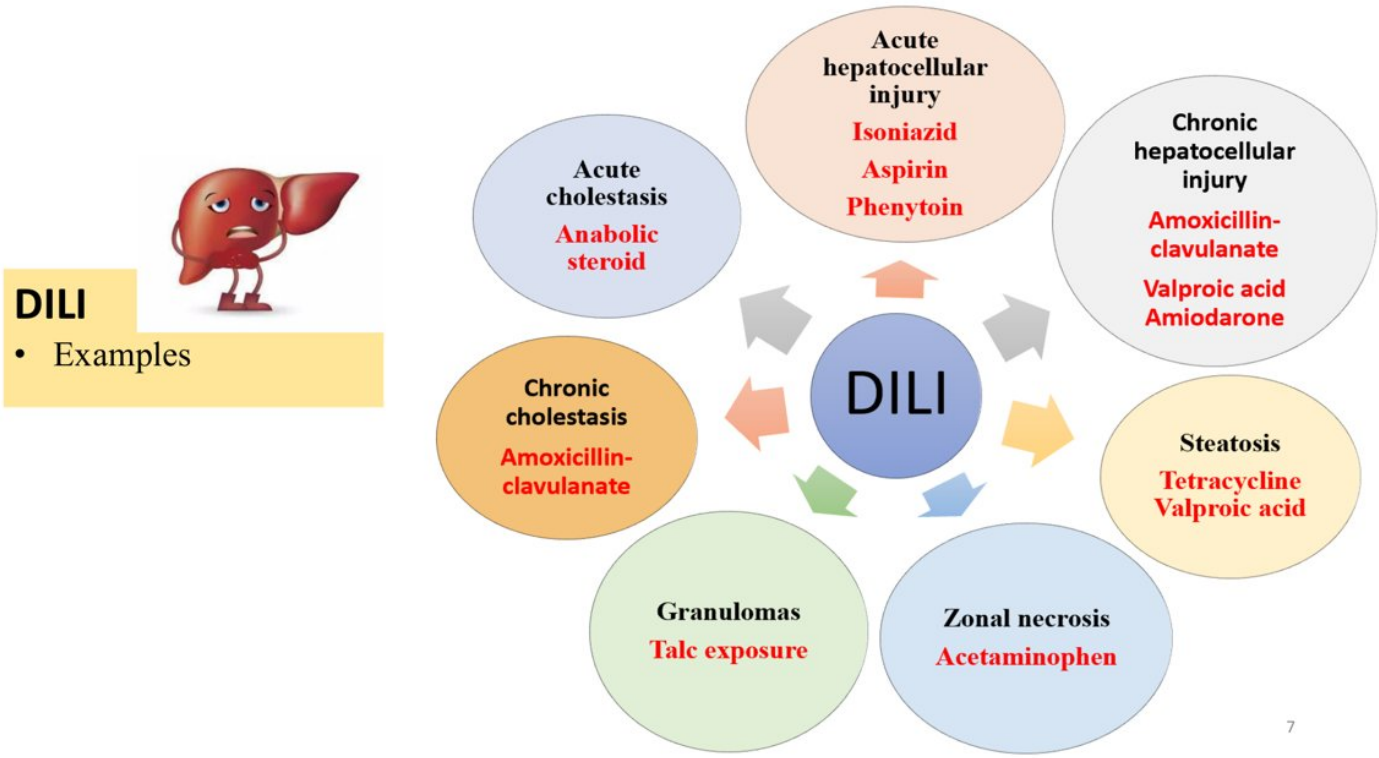
2 Classification & mechanisms

Classified by: **(1)** clinical presentation — hepatocellular / cholestatic / mixed; **(2)** mechanism of hepatotoxicity; **(3)** histological appearance on biopsy. Most DILI is asymptomatic — **jaundice** is the most common sign when present. Hepatocellular injury → ↑↑ aminotransferases; cholestatic injury → ↑ alkaline phosphatase (ALP). **Management always starts with stopping the offending drug** — prognosis is usually favorable thereafter.



Two core mechanisms: intrinsic (dose-dependent, predictable, brief latency, e.g. acetaminophen) vs idiosyncratic (unpredictable, no dose relationship), further split into hypersensitivity (immunologic, 1-5wk onset) and metabolic (aberrant metabolism, weeks-months onset) subtypes.

3 DILI patterns — drug examples



Drugs grouped by the pattern of liver injury they classically produce.

4 Key drug-specific mechanisms (case-based)

Isoniazid	Metabolized to mono-acetyl hydrazine → CYP450 converts to toxic product → hepatocellular injury (idiosyncratic, genetic susceptibility).
Aspirin (+ viral illness)	Mitochondrial dysfunction → microvesicular steatosis → **Reye's syndrome** (confusion, vomiting, ↑ammonia).
Phenytoin	Toxic CYP450 metabolite → immune-mediated hepatocellular necrosis/inflammation (hypersensitivity subtype).
Acetaminophen (overdose)	Glutathione depletion → NAPQI accumulation → hepatocellular necrosis (intrinsic, dose-dependent). Antidote: **N-acetylcysteine** — replenishes glutathione, detoxifies NAPQI directly, improves hepatic blood flow, limits oxidative stress.
Amoxicillin-clavulanate	Clavulanic acid → idiosyncratic immune-mediated cholestatic injury; can present weeks after course completion.
Tetracycline / valproic acid	Impaired mitochondrial β-oxidation (± oxidative stress/direct mitochondrial toxicity for valproate) → microvesicular steatosis.
Anabolic steroids (stanozolol)	Impaired bile transporter function → intrahepatic cholestasis; usually reversible; treat with ursodeoxycholic acid (UDCA).
Amiodarone (chronic)	Hepatic accumulation → phospholipidosis + mitochondrial dysfunction → necrosis/fibrosis over time; often only partial recovery after stopping.

5 Other drug classes & the 3-step mechanism summary

NSAIDs (sulindac)	Competitively inhibits canalicular bile salt transport → cholestatic injury (itching, jaundice).
Methotrexate	↑ plasma homocysteine → oxidative stress + proinflammatory cytokines → hepatic stellate cell activation → fibrosis.
Sulfasalazine	Usually within first month of therapy; hepatocellular or cholestatic pattern.
Azathioprine	Nodular regenerative hyperplasia (NRH) → non-cirrhotic portal hypertension.
Anti-hyperlipidemics	Hepatocellular/mixed pattern. Sustained-release **niacin** = highest hepatotoxic potential. **Statins** rarely cause significant injury (asymptomatic transaminase rise is common).

Overall 3-step DILI cascade: drugs/metabolites cause cell stress (directly or via ROS from CYP450 oxidation) → impaired mitochondrial function → triggered immune reactions.

P Must-remember pearls

- DILI is the most frequently cited reason for drug withdrawal from the market (up to 32% of withdrawals).
- Risk factors: adults > children; females > males; genetic CYP polymorphisms; obesity, diabetes, chronic viral hepatitis. African Americans: TMP-SMX is the predominant culprit; White Americans: amoxicillin-clavulanate.
- Intrinsic hepatotoxicity (e.g. acetaminophen) = dose-dependent, predictable, brief latent period, can have high mortality. Idiosyncratic = unpredictable, no dose relationship, variable latency — subdivided into hypersensitivity (1-5wk onset, rash/fever/joint pain) and metabolic (weeks-months onset).
- Hepatocellular injury → ↑↑↑ aminotransferases (AST/ALT). Cholestatic injury → ↑ALP. Treatment always starts with stopping the offending drug.
- Acetaminophen overdose: glutathione depletion → NAPQI accumulation → hepatocellular necrosis (zonal/centrilobular). Antidote = N-acetylcysteine (replenishes glutathione, direct NAPQI detox, improves hepatic blood flow, limits oxidative stress).
- Aspirin + viral illness in a child → Reye's syndrome (mitochondrial dysfunction, microvesicular steatosis, confusion, ↑ammonia) — a classic exam pairing.
- Amoxicillin-clavulanate = classic idiosyncratic immune-mediated cholestatic injury (clavulanic acid is the culprit), onset can be delayed 2+ weeks after the course ends.
- Azathioprine → nodular regenerative hyperplasia (NRH) → non-cirrhotic portal hypertension. Sulindac → cholestasis via bile salt transport inhibition. Methotrexate → fibrosis via homocysteine/stellate-cell activation. Niacin (sustained-release) has the highest hepatotoxic potential among anti-hyperlipidemics; statins rarely cause significant injury.

Q Test yourself

1. A patient develops acetaminophen-induced acute liver failure after an overdose. What is the mechanism of hepatotoxicity?

A. Depletion of hepatic glutathione leads to accumulation of the toxic metabolite NAPQI, causing hepatocellular necrosis
B. Direct inhibition of bile salt transport proteins causes cholestatic injury
C. An immune-mediated hypersensitivity reaction targets hepatocytes
D. Impairment of mitochondrial β-oxidation causes fat droplet accumulation
E. Nodular regenerative hyperplasia leads to portal hypertension

2. A patient with acetaminophen overdose is treated with N-acetylcysteine (NAC). What is NAC's primary mechanism of hepatoprotection?

A. It directly binds and inactivates acetaminophen in the stomach
B. It induces cytochrome P450 enzymes to accelerate acetaminophen clearance
C. It replenishes glutathione stores, allowing detoxification of NAPQI
D. It blocks renal excretion of the toxic metabolite
E. It stimulates hepatocyte regeneration directly

3. A child develops vomiting, confusion, and rapid breathing after taking high-dose aspirin during a viral flu-like illness. Liver biopsy shows microvesicular steatosis. What condition is this?

A. Nodular regenerative hyperplasia
B. Reye's syndrome
C. Chronic cholestasis
D. Granulomatous hepatitis
E. Zonal necrosis

4. A patient on long-term azathioprine for rheumatoid arthritis develops non-cirrhotic portal hypertension. What hepatotoxic finding is specifically associated with azathioprine?

A. Microvesicular steatosis
B. Acute cholestasis with pruritus
C. Granulomatous hepatitis
D. Nodular regenerative hyperplasia (NRH)
E. Zonal necrosis

5. A 45-year-old develops jaundice and pruritus two weeks after completing a course of amoxicillin-clavulanate (Augmentin), with elevated alkaline phosphatase and a biopsy showing cholestasis. What type of DILI mechanism does this represent?

A. An intrinsic, dose-dependent hepatotoxic reaction
B. A predictable, direct mitochondrial toxic effect
C. A reaction with a brief latent period and high mortality
D. A reaction caused by competitive inhibition of bile salt transport by the drug itself
E. An idiosyncratic, immune-mediated hypersensitivity reaction

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — Acetaminophen overdose depletes glutathione, allowing accumulation of the toxic metabolite NAPQI (N-acetyl-p-benzoquinone imine), which causes hepatocellular necrosis.

2 — C — N-acetylcysteine acts as a precursor to glutathione, replenishing hepatic glutathione stores so NAPQI can be detoxified; it also improves hepatic blood flow and limits oxidative stress.

3 — B — Aspirin use during a viral infection can cause mitochondrial dysfunction leading to Reye's syndrome — hepatocellular injury with microvesicular steatosis, elevated ammonia, confusion, and vomiting.

4 — D — Azathioprine is associated with nodular regenerative hyperplasia (NRH), which may lead to non-cirrhotic portal hypertension.

5 — E — Clavulanic acid triggers an idiosyncratic hypersensitivity reaction — the immune system recognizes clavulanic acid as foreign and attacks hepatocytes and bile duct cells, causing cholestatic injury; idiosyncratic reactions have no dose relationship and a variable latent period.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Assist. Prof. Nidhal Abdulkader & Nermin Jamal lecture slides, SGL 4 (full deck); mechanism/classification figures adapted from UpToDate; references LiverTox (NIDDK).
Study alongside the lecture, not instead of it.